

# Kent District Library Report

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## **1. Abstract**

Kent District Library (KDL) is a Kent County based library with 20 branches. They pride themselves on having a large digital library, non-book options such as video games, board games, WiFi hotspots and cameras, along with several child and adult oriented creative programs. We worked with KDL to analyze their previous expenses on the apps Overdrive and Hoopla, along with their physical collection to predict their expenses for the next few years, and to provide them with information that will be beneficial in making any needed adjustments.

## **2. Background**

### 2.1 Kent District Library

The Kent District Library system is centered in Kent County around Grand Rapids, Michigan. They have 20 locations that vary in size, with the smallest of them being a book vending machine. Their services are what one might expect from a library. Checking out books, physical audiobooks, video games, board games, electronic devices and more. They also host a plethora of other events for all ages. Kent District Library serves nearly 140,000 cardholders, a number that has been steadily rising as the area's population grows.

For our project, we worked with a team of 4 individuals from KDL: Jaci Cooper, Sheri Glon, Elizabeth Guarino and Alison Kuchta. We met on a biweekly basis, even having an in person meeting to discuss progress, our findings at the time, and any questions we had regarding their thoughts about our information.

## 2.2 Project Question

KDL came to us with something of an open-ended question. They asked for help "assessing eBook costs, circ, and trends to reexamine our collection budget allocations". With some clarification, we focused on the predictive aspects of this question. We wanted to predict future values of checkouts and needed budgets for Overdrive, Hoopla, and their physical collection. We strove to organize the information and make predictions about library usership and future budgetary needs. We focused on providing information to the team at KDL so that they can make informed decisions based on trends we found, as well as their own knowledge of the library system.

The library has three different platforms they use to check out books. They have physical books, where you go in and check out a book, and they have two online platforms to check out books right on your mobile device, Overdrive and Hoopla. KDL's physical book collection includes books, physical audiobooks, movies, and other physical media available for checkout at their branches. KDL buys roughly 12,000 physical items each month, and retires a similar amount in that same month. The library usually keeps a single book around for 5-8 years before retiring it. These retired items are then sold, donated, or added to KDL's "Little Free Library" Stock. These little free libraries are all around Kent county, and are little bookcases placed on the sides of the street by sidewalks so people can grab them as they please.

Overdrive has two apps, Libby and Kanopy. Libby has all of their ebooks, audiobooks, and magazines, and Kanopy has music, movies, and comics. Libby goes off of a digital checkout system with limited copies. Holds can be requested when all copies are checked out. KDL usually buys a new copy of a book when 6 holds are placed. Users can check out up to 40 items at any given time.

Their other digital platform Hoopla, provides ebooks, audiobooks, comics, movies, music, and magazines. It's based on a broad selection provided by Hoopla, only limited by the library's selected price. There is no maximum number of copies, but is limited by a daily budget cap. Each user is limited to 6 checkouts a month. Each book on Hoopla has a circulation cost, and checking out a book adds that cost towards the cap. KDL caps the circulation cost at \$2.99.

## 2.3 Datasets and Variables

We were given a lot of data from KDL. It mostly focused on 2021 through 2024, since those years are primarily when KDL began to utilize online services. For all platforms, we received budget information, purchase summaries, and spending data for their collections. For any information that specifies Overdrive, this includes the information from both Libby and Kanopy.

## **3. Cleaning Data Process**

There was a lot of data and spreadsheets we had to organize. We first started by sorting the spreadsheets by the platforms; Hoopla, Libby, Kanopy, and Physical books.

- [A Summary of Overdrive Purchase Information](#). This particular table was combined from multiple files.
- [Overdrive Checkout Information](#). This information is organized by genre and year.
- [Hoopla Budget Cap Blocks](#). This table shows the number of checkout blocks on Hoopla every day from 2023 and 2024.
- [Checkouts by Genre for all Platforms](#). This document details checkouts on all platforms (Overdrive, Hoopla, and Physical books) for the years we have information for them.
- [Purchase Summaries for all Platforms](#).

## 4. General Trends

### 4.1 Checkouts

After organizing the data, we began exploratory data analysis. One basic question is “what trends exist in the yearly total checkouts?” We started with checkouts from 2021 to 2024 and noticed that in general, there was an overall increase in checkouts. We also calculated the total checkouts of all the three platforms combined for each year. In the table below, the total number of checkouts is in millions, rounded to three significant figures, is shown for the three platforms: Overdrive, physical books, and Hoopla, as well as the total checkouts.

Total Checkouts for Each of the 3 Platforms in the Years 2021-2024

| Year | Overdrive | Physical | Hoopla | Total |
|------|-----------|----------|--------|-------|
| 2021 | 1.24M     | 2.73M    | 241k   | 4.21M |
| 2022 | 1.49M     | 2.75M    | 256K   | 4.50M |
| 2023 | 1.73M     | 2.80M    | 287K   | 4.81M |
| 2024 | 2.02M     | 2.77M    | 293K   | 5.08M |

Since the numbers for Hoopla are a different magnitude than those from Overdrive, hundreds of thousands vs millions, we also looked at the percentage increase for these checkout numbers in order to compare the growth across services. We calculated the percent increase of

checkouts of the current year from the previous year. We also calculated the total percent increase from 2021 to 2024. We noticed Overdrive checkouts are increasing the fastest and were surprised at how much they increased from 2021. The data is shown in the table below.

Percent change of Total Checkouts from 2022-2024

| Year                     | Overdrive | Physical | Hoopla |
|--------------------------|-----------|----------|--------|
| 2022                     | 21%       | 1%       | 6%     |
| 2023                     | 16%       | 2%       | 12%    |
| 2024                     | 17%       | -1%      | 2%     |
| Total Increase from 2021 | 63%       | 2%       | 22%    |

#### 4.2 Spending

We also wanted to look at general trends within the spending data. We initially predicted that spending would also increase because checkouts were increasing. We again started with spending from 2021 to 2024 and noticed that in general, the spending was increasing, however there were some decreases which we highlighted in red. We again calculated the total spending for all three formats combined for each year. The data is shown in the graph below.

### Total Spending for Each of the 3 Platforms in the Years 2021-2024

| Year | OverDrive | Physical Books | Hoopla | Total   |
|------|-----------|----------------|--------|---------|
| 2021 | \$1.49M   | \$1.62M        | \$450K | \$3.56M |
| 2022 | \$1.714M  | \$1.69M        | \$499K | \$3.90M |
| 2023 | \$1.710M  | \$1.83M        | \$620K | \$4.16M |
| 2024 | \$1.98M   | \$1.81M        | \$656K | \$4.48M |

We again wanted to look at the percentage increase for these spending numbers to make it easier to see how much the spending increased. We calculated the percent increase of spending of the current year from the previous year. We again highlighted the decreasing percentages in red. We also calculated the total percent increase from 2021 to 2024. We noticed spending for Hoopla is increasing the fastest and were surprised at how much it increased from 2021. The data is shown in the table below.

| Year                     | Overdrive | Physical | Hoopla |
|--------------------------|-----------|----------|--------|
| 2022                     | 15%       | 4%       | 11%    |
| 2023                     | -20%      | 8%       | 24%    |
| 2024                     | 18%       | -1%      | 6%     |
| Total Increase from 2021 | 36%       | 12%      | 46%    |



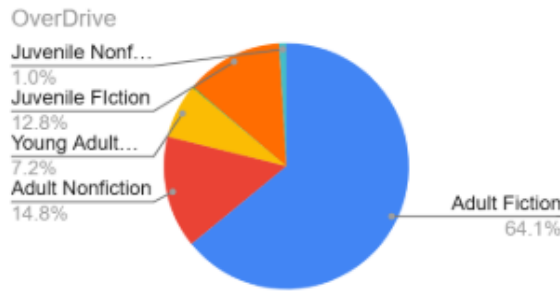
An important observation to look at here is that even though Overdrive had the biggest percent increase in checkouts Hoopla had the biggest percent increase in spending. This is because Hoopla is more expensive overall. While Overdrive experienced the largest percentage increase in checkouts between 2021 and 2024 (63%), Hoopla recorded the largest percentage increase in spending (approximately 46%). This discrepancy might be attributed to the differing pricing models employed by each platform. Overdrive operates primarily on a fixed-cost model, where the library pays once for a set number of digital licenses, which are reused by patrons over time. In contrast, Hoopla uses a usage-based pricing system, charging the library a fee for each individual checkout (up to \$2.99). As a result, even modest growth in user activity on Hoopla can lead to disproportionately higher spending. This pricing structure explains why Hoopla's budget increased more steeply than its circulation volume and underscores the importance of aligning budget planning with not just usage trends but also platform-specific cost models.

#### 4.3 Checkouts by Genre

We compared the total number of checkouts by genre on each of Libby, Hoopla, and for Physical Books. We were looking to see if any genres have grown or declined in popularity over the last few years. The graphs below show the breakdown of checkouts by genre for Libby in 2021 and 2024:

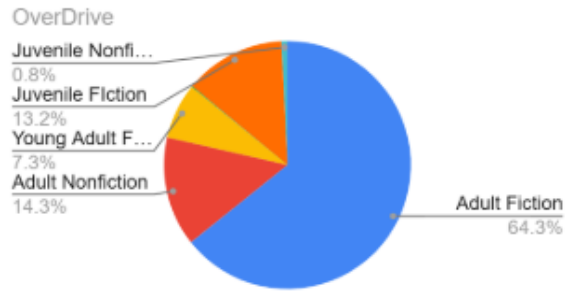
## OverDrive Checkouts by Genre

2021 - Checkouts by Genre



Total Checkouts: 1.19M

2024 - Checkouts by Genre



Total Checkouts: 1.81M

Note that in both pie charts, the Young Adult Nonfiction genre is not included. This is because the number of checkouts in this genre was around 2K in both 2021 and 2024, which is fairly insignificant when compared to the total number of checkouts represented in the graph.

These charts show that although the total number of Libby checkouts increased by about 600K from 2021 to 2024, the breakdown of the checkouts by genre stayed roughly the same. This tells us that there isn't a particular genre that is growing faster than the other; they are all growing together.

Similarly, below are the pie charts showing the breakdown of checkouts by genre for Hoopla in 2021 and 2024:

## Hoopla Checkouts by Genre

2021 - Checkouts by Genre

Hoopla

Juvenile Fiction

17.4%

Juvenile Nonfi...

2.2%

Adult Nonfiction

25.7%

Adult Fiction

54.6%

Total Checkouts: 231K

2024 - Checkouts by Genre

Hoopla

Juvenile Fiction

15.0%

Juvenile Nonfi...

1.4%

Adult Nonfiction

22.9%

Adult Fiction

60.7%

Total Checkouts: 284K

Between 2021 and 2024, the total number of Hoopla checkouts increased by about 50K, and the Adult Fiction genre grew by about 6.1%. There was about a 2.8% decline in the Adult Nonfiction genre, a 0.8% decline in Juvenile Nonfiction, and a 2.4% decline in Juvenile Fiction.

Finally, we also have the breakdown of checkouts by age group for Physical Books:

## Physical Checkouts by Age Group

2021 - Checkouts by Genre

Physical Books

Juvenile

55.7%

Adult

39.7%

Teen

4.6%

Total Checkouts: 2.66M

2024 - Checkouts by Genre

Physical Books

Juvenile

56.2%

Adult

39.4%

Teen

4.4%

Total Checkouts: 2.69M

Note that for Physical Books, we were given data on many different genres outside of Fiction and Nonfiction, so we decided to represent the data only by age group (Adult, Teen, and Juvenile).

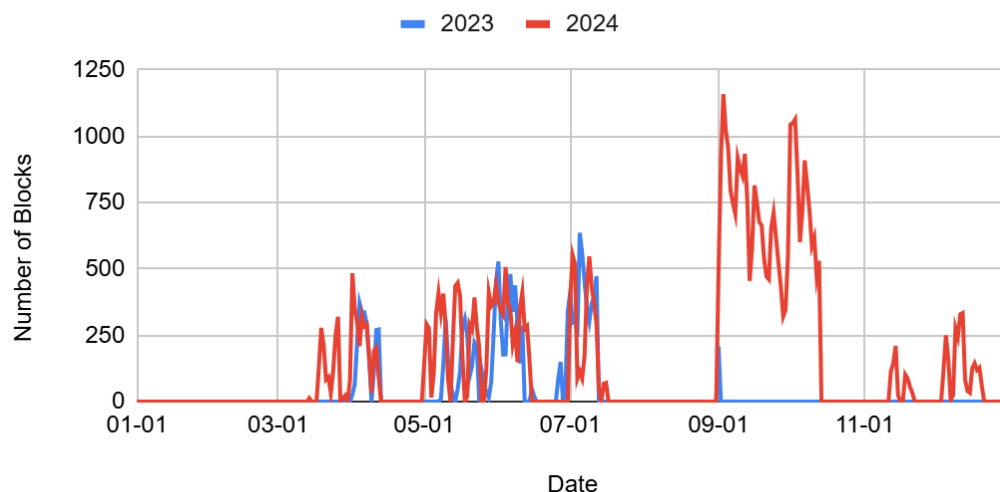
These charts show that between 2021 and 2024, the total number of checkouts increases by about 30K, and breakdown of checkouts by age group stays consistent. There isn't a particular age group that has become more or less popular over the last few years for Physical Books.

#### 4.4 Budget Cap Blocks

Each time a user checks out a title on Hoopla, there is a charge to the library. In order to regulate the amount of money KDL spends on Hoopla, there is a daily budget set aside to fund checkouts. If the budget is spent, then all users are blocked from checking out any more titles for the rest of the day. Each time a user tries to check out a title after this block occurs, one Budget Cap Block is registered with the date and time of the block. We received a spreadsheet with a list of the timestamps for all Budget Cap Blocks in 2023 and 2024. We counted the timestamps and organized the data so we could visualize the number of Budget Cap Blocks on each day in 2023 and 2024.

### Budget Cap Blocks

Hoopla



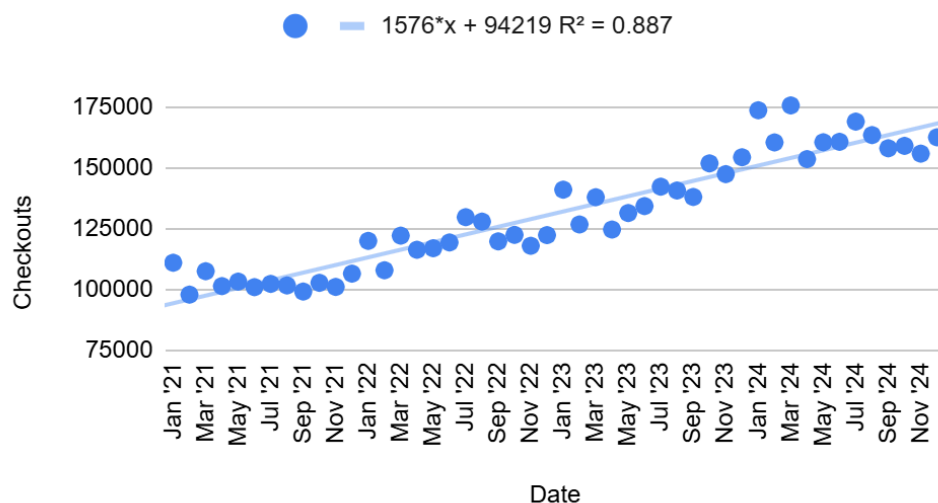
In the line graph above, the days of the year are represented along the x-axis, whereas the total number of Budget Cap Blocks occurring on each of those days is represented along the y-axis. We can observe that there are similarities between 2023 and 2024 for when spikes in blocks occur, particularly between March and July. This tells us that there is potentially a correlation between the summer months and a high traffic of checkouts, or perhaps not enough money is budgeted for this time frame. We believe that these spikes are due to seasonality. March is reading month, so it's expected that more people will read during March. Also, spring break and summer vacation allow more time for reading. We took this idea into consideration when making our predictions.

Also, note the large spike in Budget Cap Blocks in September 2024. This spike can be attributed to budgeting. In the Hoopla financing data, we noticed that the money set aside for September 2024 was \$20,000 less than the month of September in previous years. Combined with the increasing popularity of ebooks, this resulted in more checkouts than the budget could withstand, which led to a spike in Budget Cap Blocks throughout September 2024.

## **5. Linear Regression**

One of the simplest prediction models in mathematics is called a Linear Regression. Using a plot of data points over time, we can estimate an equation for a line that summarizes the trends. For example, below is a scatter plot of the checkouts on Libby during each month from 2021 to 2024:

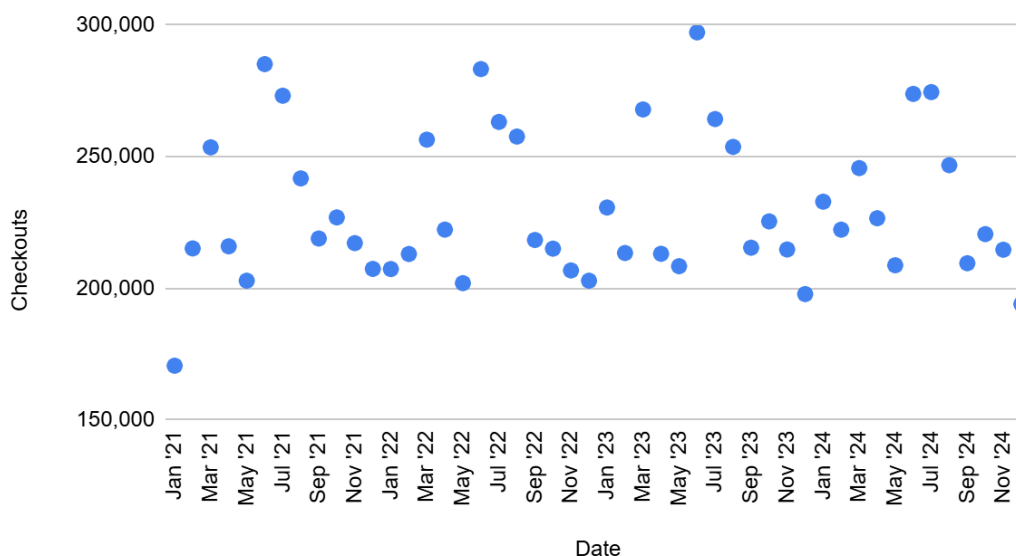
## Libby Monthly Checkout Regression



This graph shows us that between 2021 and 2024, the number of checkouts on Libby has steadily increased each month. We were able to fit a line to these data points, and the equation for this line is listed above the graph. In addition, this line has an  $R^2$  value of 0.887, which means that this line is 88.7% accurate in representing the original data points. Since this  $R^2$  value is close to 1 we could follow this line to predict checkouts for 2025, 2026, and 2027.

However, most of the data we looked at did not have a strong, linear relationship like in this example. Consider the following graph of the monthly checkouts for Physical Books:

Physical Books Monthly Checkouts



There is no linear trend in this data, so fitting a line to this graph would not give us any valuable information. One key thing to notice about this graph is the consistent increase in checkouts between March and July every year. This is the same time period when the Budget Cap Blocks increased on Hoopla, so this reinforces the idea of seasonality having an impact on checkouts each month. A Linear Regression does not take into account the nuances of seasonal trends, so we decided to use a prediction model that does. This model is called a Time Series Decomposition.

## 6. Time Series Decomposition

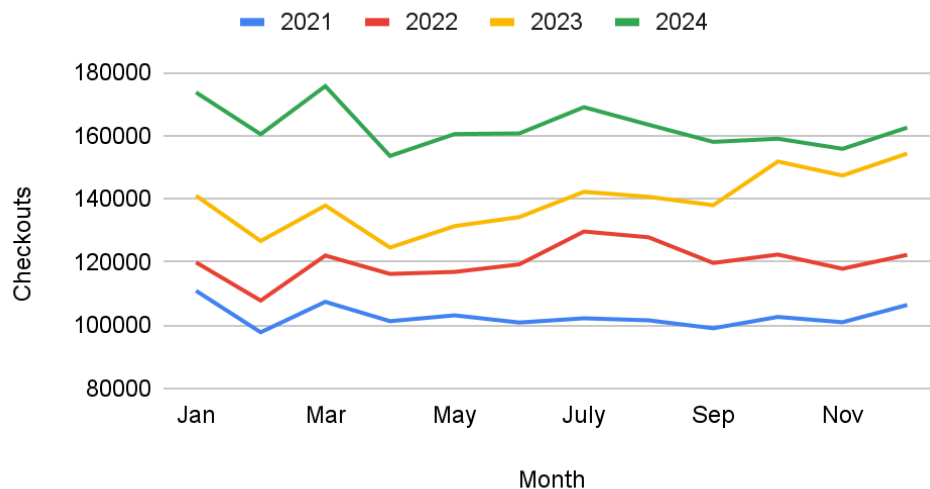
### 6.1 Definition

Time series decomposition is a data analysis technique that helps explore seasonal trends and variations within a dataset. When we were graphing the data we had, we noticed consistent

spikes in March and slightly in the summer throughout a lot of the data. This means that there is seasonality in the data. Here is the Libby Checkout data for an example.

## Checkouts by Month

Libby



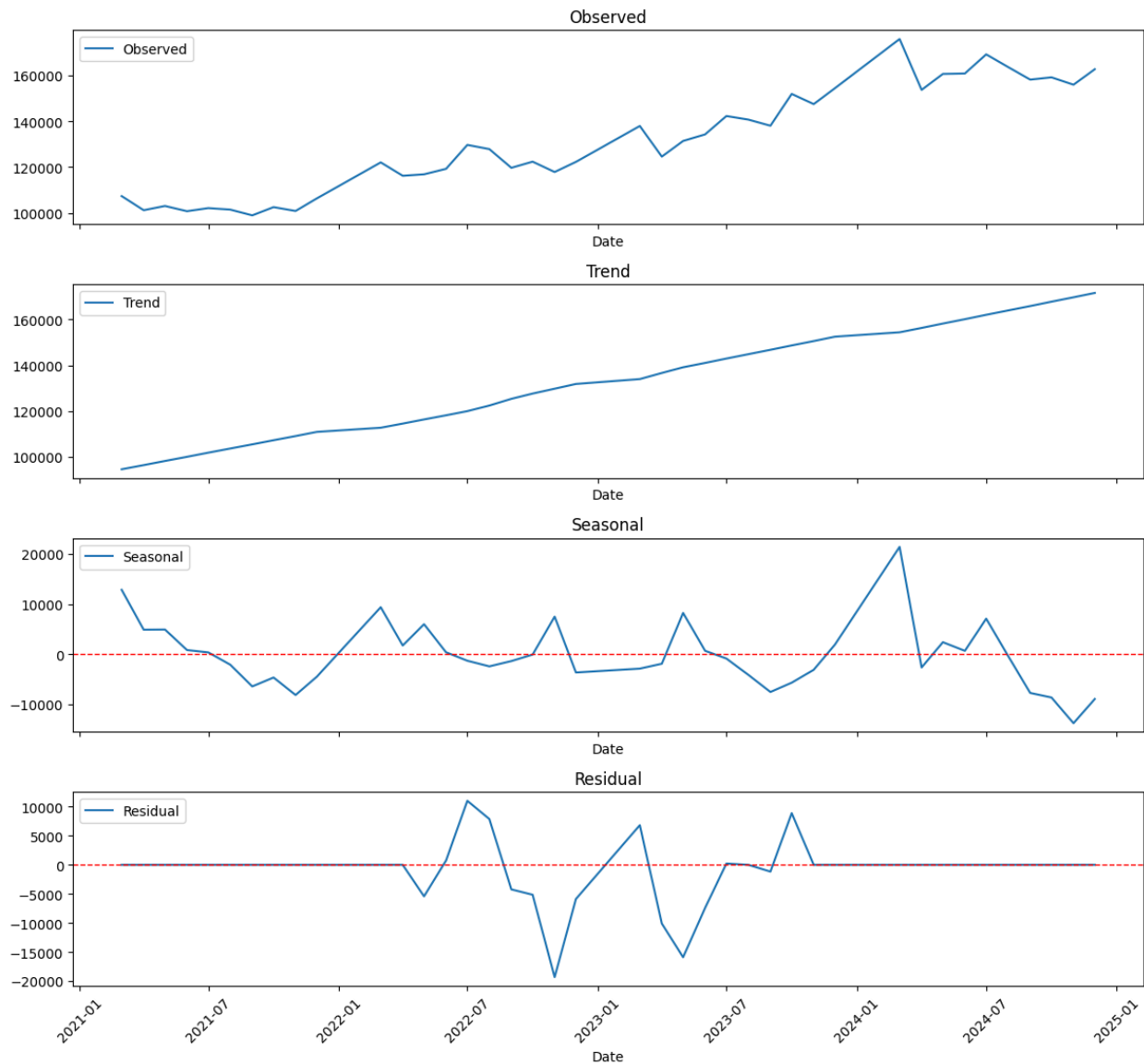
Seeing these similar bumps in the graph across each year at the same time lead us to using time series decomposition. To utilize time series decomposition, we used Python, a programming language. This Python code lets us input our dataset and output graphs describing the data, including a trend graph and a seasonality graph.

### 6.2 Analysis

We started with Overdrive's app Libby to see specific seasonality. We first wanted to see if there was any seasonality related to monthly checkouts. We organized monthly checkouts into a spreadsheet with data from January 2023 to November 2024. It was difficult to see if there were any seasonal trends throughout one year and much harder to compare years to see if they shared similar seasonal trends. Thus, we used our Python code to look at seasonal trends among both years. The data is shown in the graph below.



### Times Series Decomposition of Libby Checkouts

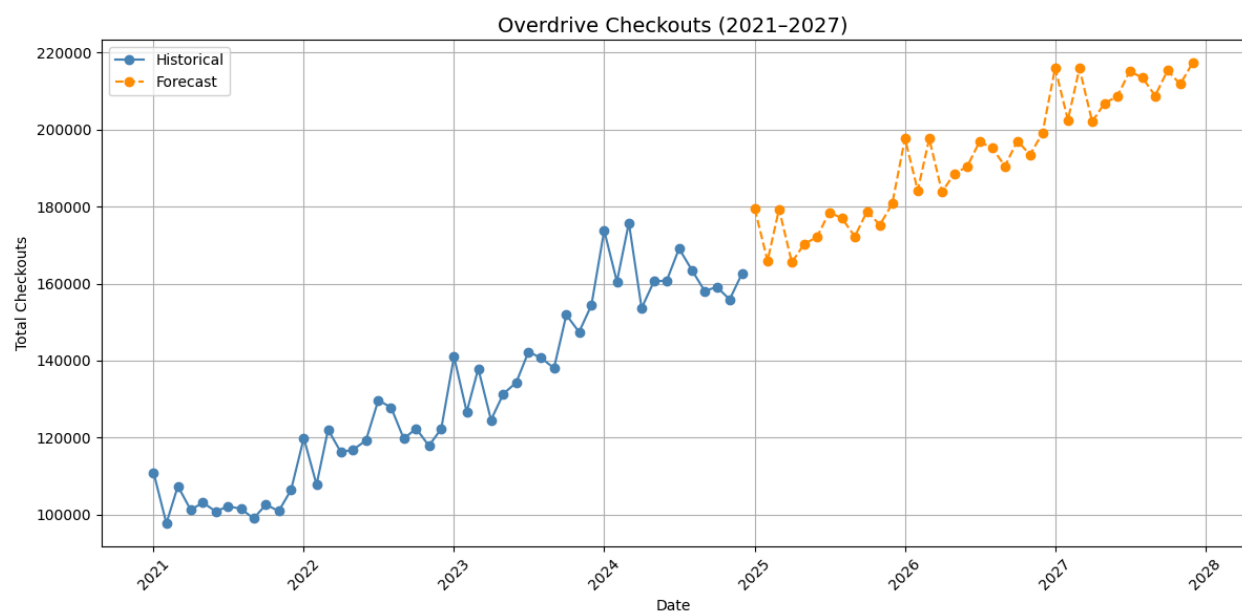


These four graphs show data for Libby checkouts from January 2021 to December 2024 focusing on total checkouts, overall trend, seasonality, and residuals. The x-axis is the months of the year from 2021 to 2024. The y-axis changes for every graph. The top graph shows the total checkouts over the four years. The y-axis is the total checkouts. The next trend graph shows the overall trend of checkouts to be increasing from January 2021 to December 2024. Again, the

y-axis is the total checkouts. Next, the seasonality graph shows any seasonal trends that show up during the four years. The y-axis is the seasonal index numbers. 0 represents a normal number of checkouts. Anything above 0 means there is an increased number of checkouts due to seasonality. Anything below 0 means there is a decrease in the number of checkouts due to seasonality. We noticed that for both years, January through July had a spike in checkouts. We concluded some factors that affected an increase in checkouts around March would be spring break and March being reading month. Finally, the residual graph is showing us that the variation in checkouts is explained by the trend graph and seasonality graph. Residual is the trend and seasonal data subtracted from the observed data

### 6.3 Predictions

We used two different forecasting models in Python to get predictions for the next three years on the cost and checkout data for Hoopla, Overdrive, and Physical books. We cleaned the data then for each data set we uploaded into the forecasting model code and got numbers for 2025, 2026, and 2027.



| Year | Physical Books | OverDrive      | Hoopla         |
|------|----------------|----------------|----------------|
| 2024 | \$1,167,198.76 | \$1,781,397.98 | \$656,068.89   |
| 2025 | \$1,670,409.19 | \$2,375,901.90 | \$881,303.40   |
| 2026 | \$1,712,244.33 | \$2,529,361.60 | \$930,095.13   |
| 2027 | \$1,816,148.63 | \$2,657,385.88 | \$1,051,205.11 |

Holt-Winters is a smoothing-based model ideal for forecasting data with a clear trend and seasonal pattern. It separates the data into three components: level (baseline), trend (upward/downward direction), and seasonality (repeating monthly or quarterly patterns). It updates these components at each step using exponentially weighted averages, putting more emphasis on recent data.

To forecast future trends in library expenditures, checkouts, and cost efficiency across digital and physical platforms, the Holt-Winters additive model was selected as the most appropriate forecasting method. Holt-Winters is particularly effective for time series data with both trend and seasonal patterns, which aligns well with library metrics that typically fluctuate based on annual usage cycles (e.g., school calendars, holiday demand).

While alternative models like SARIMA were considered, Holt-Winters offers a simpler, more interpretable structure that captures trend and seasonality directly, without requiring extensive parameter tuning. Unlike SARIMA, which models complex autoregressive structures, Holt-Winters focuses on smoothing the trend and seasonal components, making it ideal for

transparent, operational forecasting. Although SARIMA accounts for spikes, spikes are unpredictable and do not always happen so we thought the Holt-Winters Model would be a better option because it results in predicting a smooth trend for the next couple of years.

The additive variant was chosen over the multiplicative model because the seasonal changes in most datasets remained relatively constant in magnitude, rather than increasing proportionally with the overall level of the data. For example, the fluctuations in monthly OverDrive or Kanopy checkouts did not scale with total volume in a way that justified multiplicative treatment. The additive model was therefore better suited to preserve the regular seasonal shifts while avoiding exaggerated fluctuations in the forecast.

In summary, the Holt-Winters additive model provided a balance of accuracy, clarity, and robustness, making it an ideal forecasting tool for understanding and planning future library usage and budgeting trends.

## Conclusions

Our analysis of Kent District Library's physical and digital platforms from 2021 through 2024 reveals several key insights to guide future budgeting and service decisions.

### **1. Library usage for e-services is increasing, while physical book usage shows only small increases.**

As shown in the checkout trends ([Section 4.1](#)), Overdrive usage increased by 63% from 2021 to 2024, and Hoopla by 22%. In contrast, physical book checkouts increased by just 2%. This growing preference for digital platforms indicates a shift in user behavior, highlighting the need to reallocate more budget toward e-services while maintaining support for physical resources.

### **2. Genre popularity is consistent over the years.**

Our genre breakdowns ([Section 4.3](#)) show that while total checkouts have increased, the distribution of genres remains relatively stable. Whether in Hoopla, Overdrive, or physical books, no single genre is accelerating or declining in popularity. This suggests user interests are broad and balanced, and that collection development can continue focusing on variety rather than chasing trending categories.

### **3. All three platforms show seasonality.**

Time series decomposition ([Section 6](#)) revealed clear seasonal patterns in usage, especially during March (reading month), spring break, and summer vacation. These cyclical trends appeared consistently across Libby, Kanopy, and Hoopla, as well as in the spikes in Hoopla's Budget Cap Blocks ([Section 4.5](#)). Recognizing these patterns allows KDL to prepare for and manage seasonal surges more effectively, both in budgeting and in content availability.

### **4. Cost percentage increase isn't always consistent with usage increase.**

Spending trends ([Section 4.2](#)) show that increases in checkout numbers do not always align with proportional increases in spending. For instance, Hoopla's costs rose faster than its checkout rate possibly due to its per-use pricing model, while Overdrive's spending and checkouts rose more in tandem. This inconsistency emphasizes the importance of understanding each platform's pricing structure when planning budget allocations.

Ultimately, the Holt-Winters additive model provided reliable forecasts ([Section 6.3](#)) for future usage and spending, thanks to its ability to capture both long-term trends and seasonal variation. While SARIMA models were considered, Holt-Winters offered clearer interpretation with less complexity—an important factor for operational planning.

We recommend that KDL continue tracking seasonal usage, monitor cost-per-circulation across platforms, and consider shifting more budget resources toward digital services while

maintaining support for physical materials that still serve a steady, if slower-growing, base of users.

## Additional Charts and Tables

 Prediction Graphs

 Prediction Summary Charts

 Times Series Decomposition Graphs

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